

Shuqi Fu

Ph.D. Student in Astrophysics

Kavli Institute for Astronomy and Astrophysics, Peking University

fushuqi@stu.pku.edu.cn | (+86) 18859088166 | ORCID: 0000-0003-0964-7188

Homepage: <https://shuqifu2000.github.io>

Research Interests

High-redshift galaxy formation and evolution; cosmic star-formation history; the Epoch of Reionization; early black-hole growth and galaxy-AGN coevolution; JWST emission-line spectroscopy and wide-field surveys.

Education

Peking University, Beijing, China

Sep 2022 – Jul 2027 (expected)

Ph.D. in Astrophysics, Kavli Institute for Astronomy and Astrophysics (KIAA)

Advisor: Prof. Linhua Jiang

The University of Manchester, Manchester, UK

Feb 2026 – Feb 2027

Visiting Ph.D. Student, Jodrell Bank Centre for Astrophysics; Host: Prof. Christopher J. Conselice

University of Science and Technology of China, Hefei, China

Sep 2018 – Jul 2022

B.S. in Physics; Advisor: Prof. Junxian Wang

Undergraduate thesis: *A Spectroscopic Survey of Lyman-break Galaxies at $z \sim 6$*

Publications and Manuscripts

First- and Second-Author Papers

- [1] **Fu, S.**, Zhang, Z., Jiang, D., Chen, J., Jiang, L., Ho, L. C., Inayoshi, K., et al. (2026), “The discovery of two little red dots in transition into quasars,” *Nature Astronomy*, 10, 1372–1386.
- [2] **Fu, S.**, et al. (2026), “A Comprehensive JWST Slitless Spectroscopic Census of $\sim 12,000$ Emission-line Galaxies at $0 < z < 9$ in COSMOS,” submitted to *The Astrophysical Journal Supplement Series*.
- [3] **Fu, S.**, Conselice, C. J., et al. (2026), “The Ionizing Photon Budget of $H\alpha$ -emitting Galaxies at $z \sim 5$ –6: Production Efficiency, Escape Fraction, and Galaxy Morphology,” manuscript in preparation.
- [4] **Fu, S.**, et al. (2026), “A JWST View of Cosmic Star Formation from $z \sim 0$ to 9 with Hydrogen Recombination Lines,” manuscript in preparation.
- [5] **Fu, S.**, Sun, F., Jiang, L., et al. (2025), “Medium-band Astrophysics with the Grism of NIRC2 in Frontier Fields (MAGNIF): Spectroscopic Census of $H\alpha$ Luminosity Functions and Cosmic Star Formation at $z \sim 4.5$ and 6.3,” arXiv:2503.03829; submitted to *The Astrophysical Journal*.
- [6] Jiang, L., **Fu, S.**, et al. (2025), “Constraints on the Variation of the Fine-structure Constant at $3 < z < 10$ with JWST Emission-line Galaxies,” *The Astrophysical Journal*, 980, 93.
- [7] **Fu, S.**, Jiang, L., Ning, Y., Liu, W., & Pan, Z. (2024), “The Magellan M2FS Spectroscopic Survey of High-redshift Galaxies: The Brightest Lyman-break Galaxies at $z \sim 6$,” *The Astrophysical Journal*, 963, 51.
- [8] **Fu, S.**, Brandt, W. N., Zou, F., Laor, A., et al. (2022), “The Nature of Luminous Quasars with Very Large C IV Equivalent Widths,” *The Astrophysical Journal*, 934, 97.

Other Co-authored Publications

- [1] Jiang, D., Jiang, L., **Fu, S.**, et al. (2026), “COSMOS-3D: Black Hole Mass Estimators and Luminosity Functions of Paschen-line AGNs,” *The Astrophysical Journal*, 1004, 92.
- [2] Li, Z.-J., Zou, S., Lyu, J., Champagne, J. B., Huang, J.-S., Cheng, C., **Fu, S.**, et al. (2026), “COSMOS-3D: Dense Circumnuclear Gas Across Black Hole Growth Phases at $z \sim 3$,” *The Astrophysical Journal*, 1002, 9.
- [3] Ning, Y., Cai, Z., Jiang, L., Lin, X., **Fu, S.**, & Spinoso, D. (2023), “An $H\alpha$ Impression of $Ly\alpha$ Galaxies at $z \simeq 6$ with Deep JWST/NIRCam Imaging,” *The Astrophysical Journal Letters*, 944, L1.
- [4] Kang, J.-L., Wang, J.-X., & **Fu, S.** (2023), “What Can Be Learnt from a Highly Informative X-ray Occultation Event in NGC 6814? A Marvellous Absorber,” *Monthly Notices of the Royal Astronomical Society*, 525, 1941–1952.

Teaching

- Teaching Assistant, Fundamental Astronomy, Peking University, 2023
- Teaching Assistant, Thermal Physics, University of Science and Technology of China, 2021

Honors and Scholarships

- State-Sponsored Overseas Research Scholarship for Outstanding Ph.D. Students, China Scholarship Council, 2026–2027
- National Scholarship, Ministry of Education of the People’s Republic of China, 2025

Selected Conferences

- **Kavli Astrophysics Symposium**, Cambridge, UK 2026
Talk: *Tracing Star Formation Across Cosmic Time with JWST*
- **EREBUS / COSMOS-3D Meeting**, Ann Arbor, MI, USA 2026
Talk: *A Large Sample of $\sim 12,000$ Emission-line Galaxies in COSMOS and the Discovery of Two LRDs in Transition into Quasars*
- **Euclid-UK and LSST:UK Meeting**, Manchester, UK 2026
- **EREBUS / COSMOS-3D Meeting**, Bologna, Italy 2025
Talk: *JWST/MAGNIF: $H\alpha$ Luminosity Functions to Unprecedentedly Faint Limits beyond Cosmic Noon*
- **The 27th Guo Shoujing Symposium and 2025 Workshop on Frontiers in Galaxy Cosmology**, Changsha, China 2025
Talk: *JWST/MAGNIF: $H\alpha$ Luminosity Functions to Unprecedentedly Faint Limits beyond Cosmic Noon*
- **Probing Supermassive Black Holes**, Kashiwa, Japan 2024
- **2023 CSST Workshop on Key Questions in Galaxy Science**, Anqing, China 2023
Talk: *Spectroscopic Properties of Lyman-break Galaxies at $z \sim 6$*

Research Skills

Observations	JWST/NIRCam imaging and WFSS; JWST/NIRSpec and Magellan/M2FS spectroscopy; HST, Subaru/HSC, SDSS, and Chandra data analysis
Methods	Slitless spectroscopic data reduction and forward modeling; spectral analysis and emission-line measurements; SED fitting; injection–recovery completeness simulations; non-parametric galaxy morphology; maximum-likelihood estimation and Bayesian inference (MCMC)
Computing	Python, Linux, MATLAB, and C++